MSG No. 53.332 P_cmna [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2/m..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,5,6]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,4,7,8]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[9,10,13,14]`
4	$\left[\frac{1}{2}, 0, 0\right]$	`[11,12,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `2'/m..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,6,10,13]`
2	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[2,5,9,14]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[3,8,12,15]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[4,7,11,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,5,6]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,4,7,8]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,13,14]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[11,12,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2'/m..`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,6,10,13]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[2,5,9,14]`
3	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[3,8,12,15]`
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[4,7,11,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, 0\right]$	`[1,2,11,12]`
2	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[3,4,9,10]`
3	$\left[\frac{3}{4}, 0, 0\right]$	`[5,6,15,16]`
4	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[7,8,13,14]`

Table 6: Wyckoff site: `4f`, site symmetry: `2'22'`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	`[1,3,10,12]`
2	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[2,4,9,11]`
3	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	`[5,7,14,16]`
4	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[6,8,13,15]`

Table 7: Wyckoff site: `4g`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,2,11,12]`
2	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,4,9,10]`
3	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[5,6,15,16]`
4	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,8,13,14]`

Table 8: Wyckoff site: `4h`, site symmetry: `2'22'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,3,10,12]`
2	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[2,4,9,11]`
3	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[5,7,14,16]`
4	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[6,8,13,15]`

Table 9: Wyckoff site: `8i`, site symmetry: `..2'`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, z\right]$	`[1,12]`
2	$\left[\frac{1}{4}, 0, -z\right]$	`[2,11]`
3	$\left[\frac{1}{4}, 0, \frac{1}{2} - z\right]$	`[3,10]`
4	$\left[\frac{1}{4}, 0, z + \frac{1}{2}\right]$	`[4,9]`
5	$\left[\frac{3}{4}, 0, -z\right]$	`[5,16]`
6	$\left[\frac{3}{4}, 0, z\right]$	`[6,15]`
7	$\left[\frac{3}{4}, 0, z + \frac{1}{2}\right]$	`[7,14]`
8	$\left[\frac{3}{4}, 0, \frac{1}{2} - z\right]$	`[8,13]`

Table 10: Wyckoff site: `8j`, site symmetry: `..2'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, z\right]$	`[1,12]`
2	$\left[\frac{1}{4}, \frac{1}{2}, -z\right]$	`[2,11]`
3	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[3,10]`
4	$\left[\frac{1}{4}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[4,9]`
5	$\left[\frac{3}{4}, \frac{1}{2}, -z\right]$	`[5,16]`
6	$\left[\frac{3}{4}, \frac{1}{2}, z\right]$	`[6,15]`

continued ...

Table 10

No.	position	mapping
7	$\left[\frac{3}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[7,14]
8	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[8,13]

Table 11: Wyckoff site: 8k, site symmetry: .2'.

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ 0\right]$	[1,11]
2	$\left[\frac{1}{4},\ -y,\ 0\right]$	[2,12]
3	$\left[\frac{1}{4},\ y,\ \frac{1}{2}\right]$	[3,9]
4	$\left[\frac{1}{4},\ -y,\ \frac{1}{2}\right]$	[4,10]
5	$\left[\frac{3}{4},\ -y,\ 0\right]$	[5,15]
6	$\left[\frac{3}{4},\ y,\ 0\right]$	[6,16]
7	$\left[\frac{3}{4},\ -y,\ \frac{1}{2}\right]$	[7,13]
8	$\left[\frac{3}{4},\ y,\ \frac{1}{2}\right]$	[8,14]

Table 12: Wyckoff site: 8l, site symmetry: .2.

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ \frac{1}{4}\right]$	[1,3]
2	$\left[\frac{1}{4},\ -y,\ \frac{3}{4}\right]$	[2,4]
3	$\left[\frac{3}{4},\ -y,\ \frac{3}{4}\right]$	[5,7]
4	$\left[\frac{3}{4},\ y,\ \frac{1}{4}\right]$	[6,8]
5	$\left[\frac{1}{4},\ y,\ \frac{3}{4}\right]$	[9,11]
6	$\left[\frac{1}{4},\ -y,\ \frac{1}{4}\right]$	[10,12]
7	$\left[\frac{3}{4},\ -y,\ \frac{1}{4}\right]$	[13,15]
8	$\left[\frac{3}{4},\ y,\ \frac{3}{4}\right]$	[14,16]

Table 13: Wyckoff site: 8m, site symmetry: 2..

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,2]
2	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[3,4]
3	$[-x,\ 0,\ 0]$	[5,6]
4	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[7,8]
5	$\left[x,\ 0,\ \frac{1}{2}\right]$	[9,10]
6	$\left[\frac{1}{2}-x,\ 0,\ 0\right]$	[11,12]
7	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[13,14]
8	$\left[x+\frac{1}{2},\ 0,\ 0\right]$	[15,16]

Table 14: Wyckoff site: `8n`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,10]`
2	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[2,9]`
3	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[3,12]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[4,11]`
5	$\left[-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,14]`
6	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[6,13]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[7,16]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[8,15]`

Table 15: Wyckoff site: `8o`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,2]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,4]`
3	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[5,6]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[7,8]`
5	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	`[11,12]`
7	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[13,14]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[15,16]`

Table 16: Wyckoff site: `8p`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,10]`
2	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[2,9]`
3	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{4}\right]$	`[3,12]`
4	$\left[\frac{1}{2}-x,\ 0,\ \frac{3}{4}\right]$	`[4,11]`
5	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[5,14]`
6	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[6,13]`
7	$\left[x+\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[7,16]`
8	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[8,15]`

Table 17: Wyckoff site: `8q`, site symmetry: `m..`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,6]`
2	$[0,\ -y,\ -z]$	`[2,5]`
3	$\left[\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	`[3,8]`
4	$\left[\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	`[4,7]`
5	$\left[0,\ y,\ z+\frac{1}{2}\right]$	`[9,14]`
6	$\left[0,\ -y,\ \frac{1}{2}-z\right]$	`[10,13]`
7	$\left[\frac{1}{2},\ y,\ -z\right]$	`[11,16]`

continued ...

Table 17

No.	position	mapping
8	$\left[\frac{1}{2}, -y, z\right]$	`[12,15]`

Table 18: Wyckoff site: `16r`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$[x, -y, -z]$	`[2]`
3	$\left[\frac{1}{2} - x, y, \frac{1}{2} - z\right]$	`[3]`
4	$\left[\frac{1}{2} - x, -y, z + \frac{1}{2}\right]$	`[4]`
5	$[-x, -y, -z]$	`[5]`
6	$[-x, y, z]$	`[6]`
7	$\left[x + \frac{1}{2}, -y, z + \frac{1}{2}\right]$	`[7]`
8	$\left[x + \frac{1}{2}, y, \frac{1}{2} - z\right]$	`[8]`
9	$\left[x, y, z + \frac{1}{2}\right]$	`[9]`
10	$\left[x, -y, \frac{1}{2} - z\right]$	`[10]`
11	$\left[\frac{1}{2} - x, y, -z\right]$	`[11]`
12	$\left[\frac{1}{2} - x, -y, z\right]$	`[12]`
13	$\left[-x, -y, \frac{1}{2} - z\right]$	`[13]`
14	$\left[-x, y, z + \frac{1}{2}\right]$	`[14]`
15	$\left[x + \frac{1}{2}, -y, z\right]$	`[15]`
16	$\left[x + \frac{1}{2}, y, -z\right]$	`[16]`